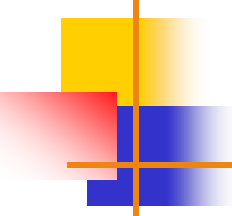


Data Mining for Business Analytics



Learning Objectives

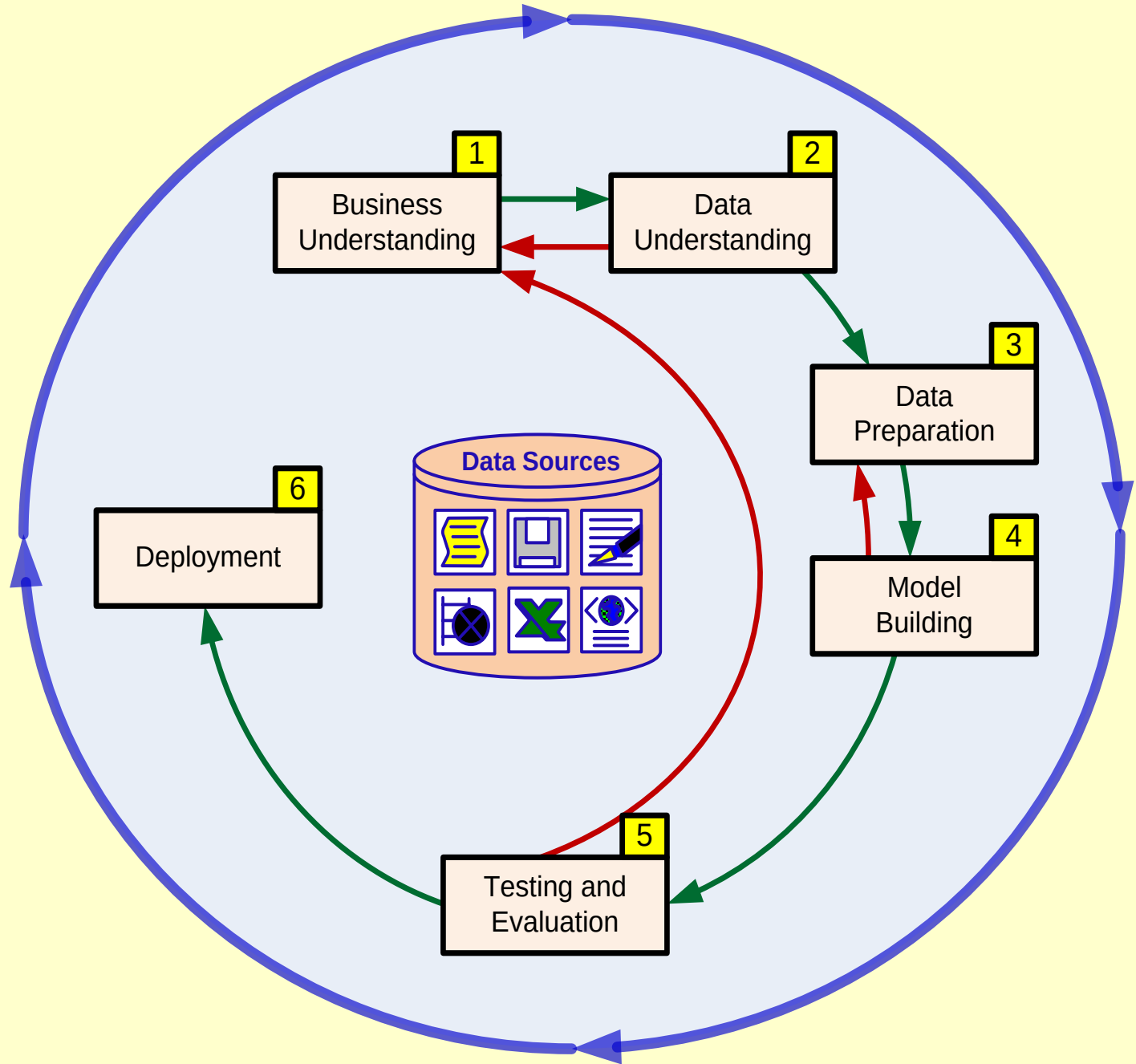
- Understand the objectives and benefits of machine learning as a technique of Data mining (Business Analytics)
- Understand the process of data mining
- Learn selected methods and algorithms of Clustering as a Data mining technique



What Does Data Mining (DM) Do?

- DM extract patterns from data
 - Pattern = A mathematical (numeric and/or symbolic) relationship among data items
- Types of patterns (Machine Learning model)
 - Supervised Learning for prediction
 - Classification e.g. spam vs. not spam
 - Regression e.g. stock prices
 - Unsupervised Learning for grouping similar attributes
 - Cluster e.g. customer segmentation
 - Association Mining e.g. market basket analysis
 - Dimensionality reduction e.g. reduce attributes for modeling

Data Mining Process: CRISP-DM

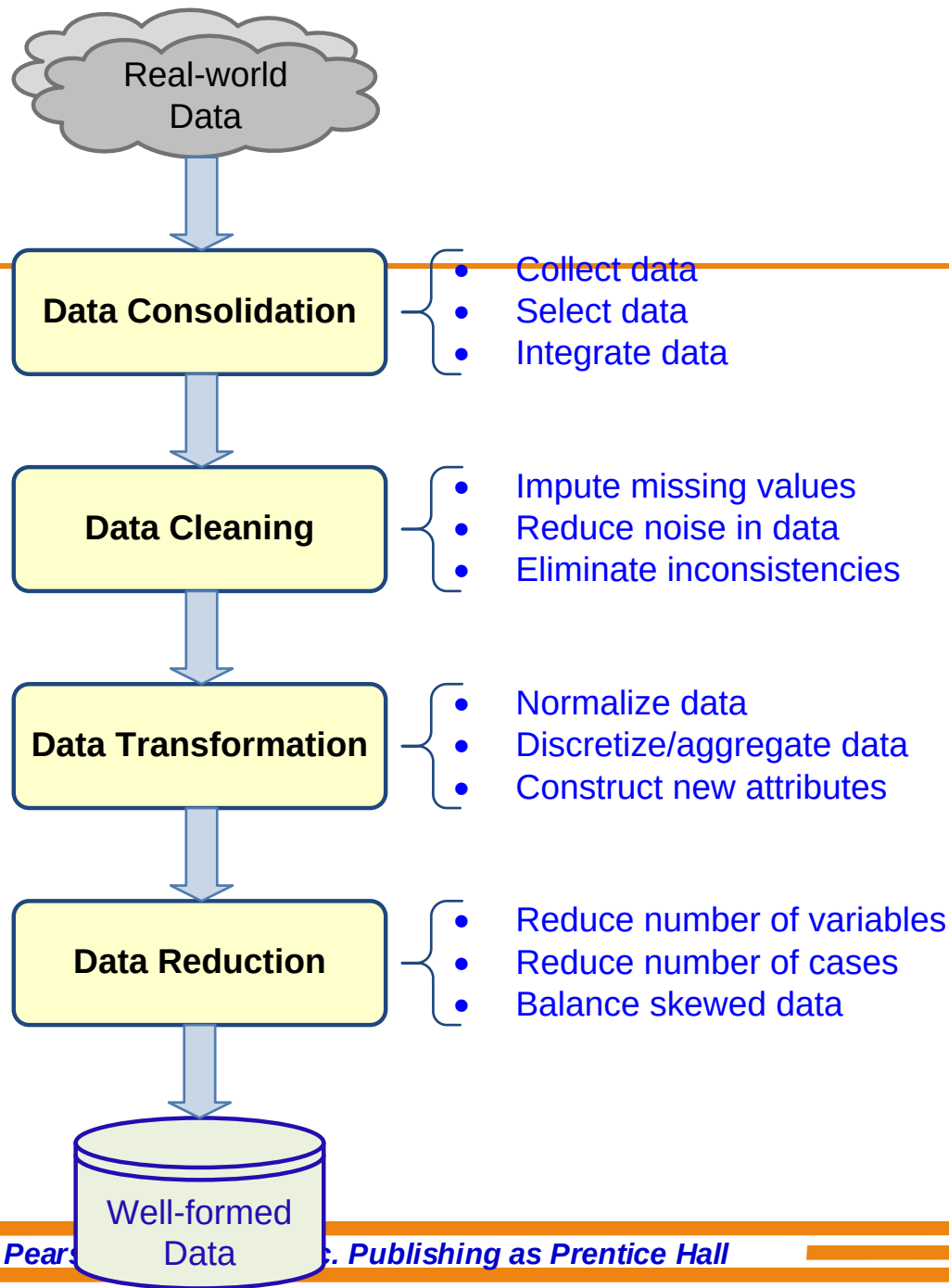




Data Mining Overview

- **Business understanding:** situational context, specific objectives, deliverables
- **Data understanding:** collecting raw data, preliminary results, potential hypotheses
- **Data preparation:** records and variable selection, wrangling, cleaning
- **Modeling:** selection and execution of data mining techniques, convert or transform data to formats/types needed by certain analyses or assumptions, validation
- **Evaluation:** evaluate performance of competing models, select best models, review and interpret results, develop recommendations
- **Deployment:** develop a set of actionable insights and a strategy for deployment/monitoring/feedback

Data Preparation – A Critical DM Task

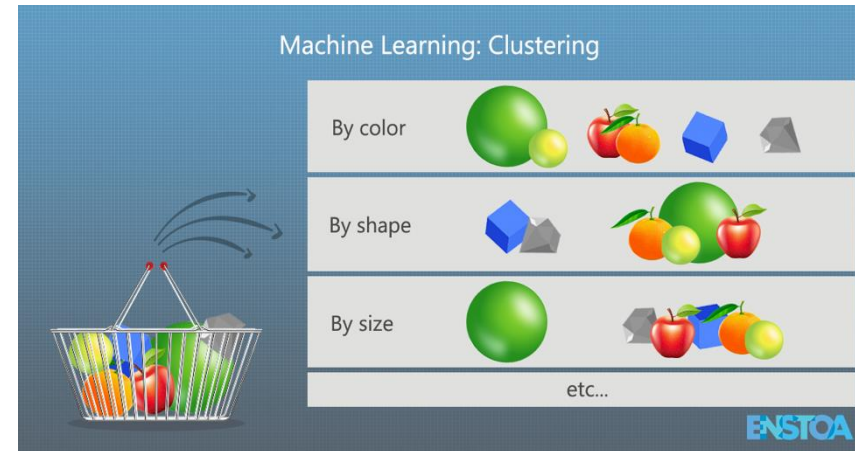


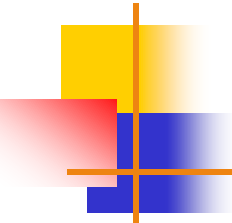
Clustering: Unsupervised machine Learning

- Used for automatic identification of natural groupings of things
 - May used to generate label (Prior to supervised learning)
- Employ unsupervised learning
 - Also known as segmentation

- Example:

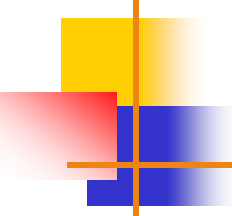
- Reduce variables from IoT's devices which tend to have a lot of redundant data.
- Segment customers into groups for customized product and service offerings





Cluster Analysis

- Example of available methods
 - **Algorithmic and Statistical methods** (k-mean, Hierarchical Clustering)
 - **Neural networks** (adaptive resonance theory [ART], Self-Organizing Map [SOM])
 - **Association mining** (Apriori, Frequent Pattern Growth)



Cluster Analysis for Data Mining

- Most Statistical cluster analysis methods involve the use of a **distance or similarity measure** to calculate the closeness between pairs of items
 - Usually, the “**Minkowski distance**”
$$\left(\sum_{i=1}^r (a_i + b_i)^p \right)^{1/p}$$
- Based on the distance between observations
 - Small distance implies high similarity
 - Large distance implies low similarity

Similarity Measures for numerical data

- Suppose we have k numerical variables $x_1, x_2, \dots, x_k$

- Euclidean distance (P=2)

- Widely used
- Distance between two observations i and j

$$\left(\sum_{i=1}^r (a_i + b_i)^p \right)^{1/p}$$

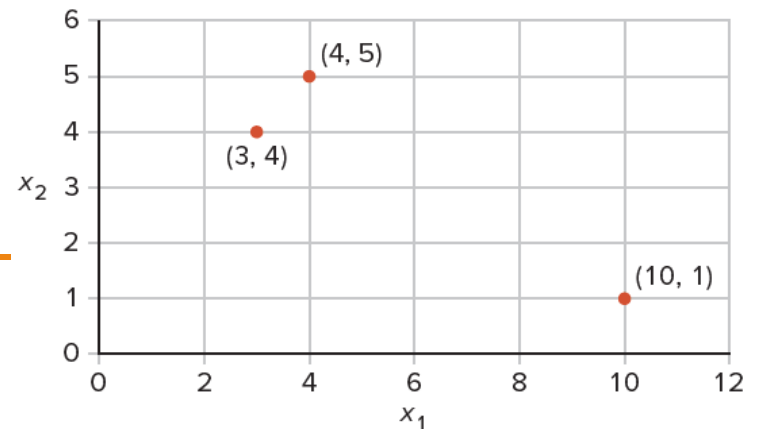
- $\sqrt{(x_{1i} - x_{1j})^2 + (x_{2i} - x_{2j})^2 + \dots + (x_{ki} - x_{kj})^2}$

- Manhattan distance (P=1)

- Shortest path for a vehicle to follow
- Shortest distance horizontally/vertically between observations i and j
- $|x_{1i} - x_{1j}| + |x_{2i} - x_{2j}| + \dots + |x_{ki} - x_{kj}|$

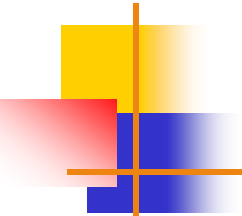
Similarity Measures

■ Example



Observation	x_1	x_2
1	3	4
2	4	5
3	10	1

- Calculate and interpret the Euclidean distance.
- Calculate and interpret the Manhattan distance.



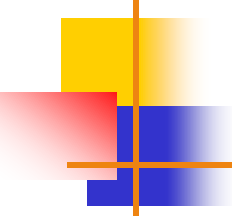
Similarity Measures

Euclidean

- 1 and 2: $\sqrt{(x_{1i} - x_{1j})^2 + (x_{2i} - x_{2j})^2} = \sqrt{(3 - 4)^2 + (4 - 5)^2} = 1.41$
- 1 and 3: 7.62
- 2 and 3: 7.21
- Interpretation: 1 and 2 are the most similar

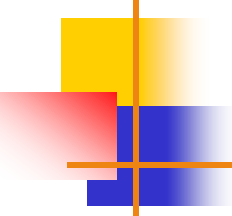
Manhattan

- 1 and 2: $|x_{1i} - x_{1j}| + |x_{2i} - x_{2j}| = |3 - 4| + |4 - 5| = 2$
- 1 and 3: 10
- 2 and 3: 10
- Interpretation: 1 and 2 are the most similar, however, take note on the ability of Euclidean to differentiate the distance of 1-3 from 2-3



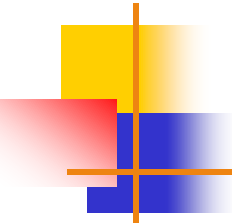
Clustering: The K-Means Algorithm

1. Choose K , the number of clusters.
2. Randomly choose K points as cluster centers.
3. Assign the remaining instances to their **closest** cluster Center or Centroid, e.g.,
 - Based on Minkowski distance to the center.
4. Calculate a new cluster center for each cluster, e.g.,
 - Average values of all member in the cluster
5. Repeat steps 2-4 until the cluster centers do not change (Convergence).



Note for generating the new center of clusters.

- In step 4, calculating a new cluster center (or centroid) for each cluster (C_i)
 - Centroid (ct_i) is defined as the weighted summation of all objects (x) in the cluster C_i .
 - Can be represented in math form:
 - $ct_i = \frac{\sum_{x \in C_i} x}{|C_i|}$

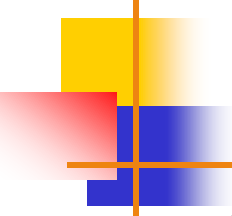


Clustering performance

- **Clustering** with K clusters, $K=1,\dots,J$ where each cluster has a center μ_k , contains data points x_i , can be measured for its performance using:

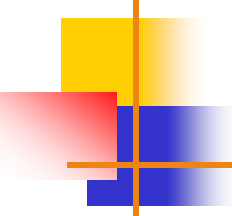
$$\text{RSS} = \sum_{k=1}^K \sum_{i \in C_k} \|\mathbf{x}_i - \mu_k\|^2$$

- $\|x_i - \mu_k\|^2$ = Squared Euclidean distance between a point and the cluster centroid
- It quantifies how tightly the data points in a cluster are grouped around the cluster centroid.
 - The tighter the better



Finding the optimal K

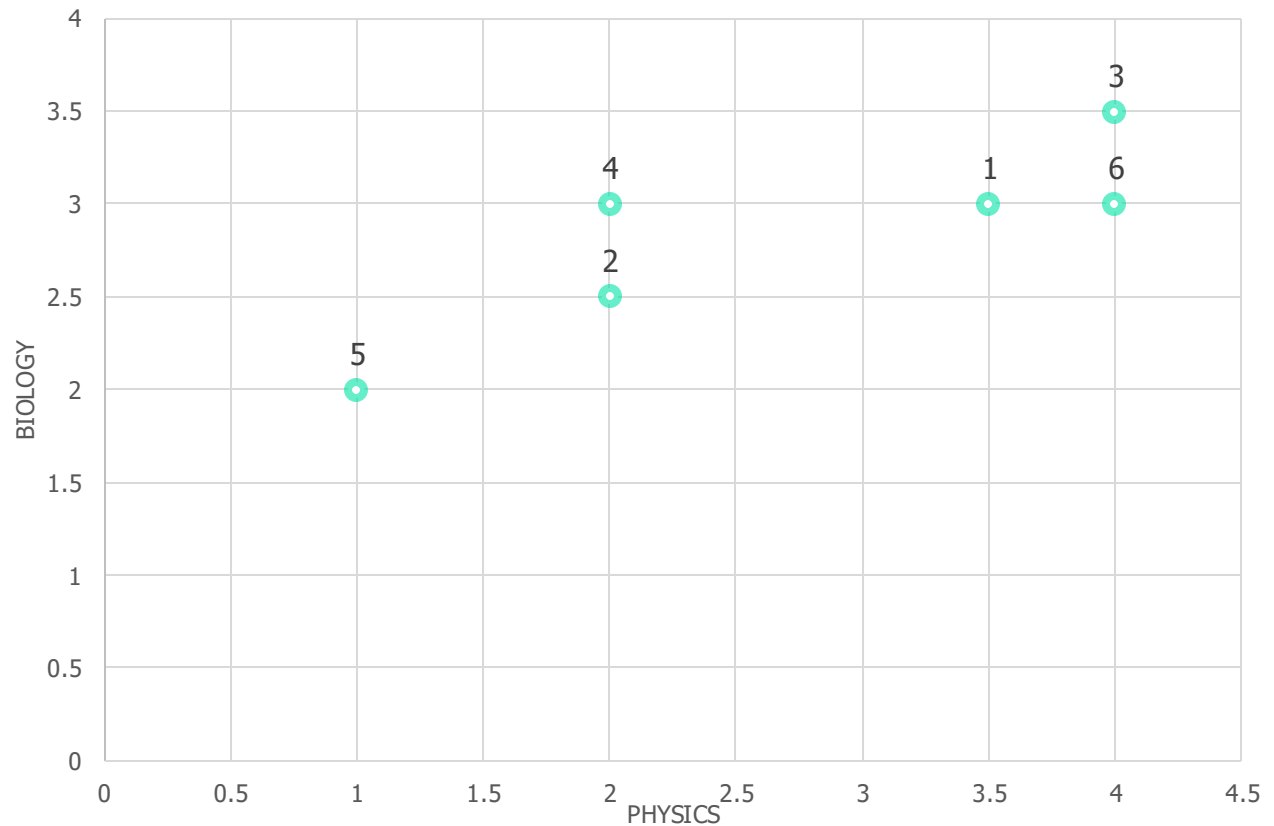
- **Choosing k :** Use the **Elbow Method** or **Silhouette Score** to find the optimal k .
- **Scalability:** K-Means works well for large datasets but may struggle with very high dimensions.
- **Limitations:** Sensitive to initial centroids and cannot handle non-convex clusters well.

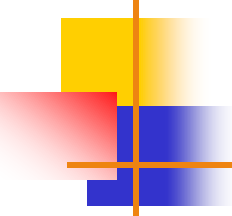


Elbow Method – Finding the Optimal k

- The **Elbow Method** evaluates how the **RSS (Residual Sum of Squares)** or **Inertia** changes as you increase k .
- The optimal k is at the "elbow point," where additional clusters provide diminishing returns in reducing inertia.

An example of K-Means





General Considerations

- Requires *real-valued* data.
- We must select the number of clusters present in the data.
 - There is no “truly optimal” way to calculate it
 - Practically, start from 1 clusters until obtain the best RSS, or
 - Heuristics rule.
 - Look at the sparseness of clusters
 - Number of clusters = $(n/2)^{1/2}$ (n: no of data points)
- Attribute significance cannot be determined.



Objectives Review

- Data mining = process of identifying patterns in data stored in structured databases
- Unsupervised learning
 - Similarity measure (Minkowsky distance)
 - Clustering, sample method = k-mean



Further readings

- K-mean clustering

http://home.deib.polimi.it/matteucc/Clustering/tutorial_html/kmeans.html